

Comprehensive Org mode syntax and features reference

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This document demonstrates a broad range of Org mode features. It serves as a comprehensive reference for writing technical documentation, producing HTML and PDF output, embedding source code, and testing CSS, $\LaTeX$, and export behavior.

Org mode provides a simple *plain-text* markup language for creating well-structured documents. From a single source file, you can produce high-quality HTML, $\LaTeX$, PDF, and many other output formats while maintaining an efficient writing workflow.

1. Basics

This chapter introduces the fundamental building blocks of an Org document.

1.1. Biggest heading

Start a new chapter by creating a top-level heading.

1.1.1 Bigger heading

Use section headings to organize related content.

Big heading

Subsections help divide longer sections into manageable parts.

Paragraphs and line breaks

A single newline does not create a new paragraph. The following lines belong to the same paragraph.

But an empty line

demarcates a new paragraph.

To force a line break without beginning a new paragraph, end a line with 2 backslashes. The next line will appear immediately below it.

To insert a horizontal rule, write at least 5 consecutive hyphens: this text appears above the horizontal rule.

and this text appears below it.

Numbered headings

You can automatically number headings up to a specified depth by setting the appropriate export option.

```
#+OPTIONS: H:4
```

1.1.2 Text width

The following paragraph is intentionally long to demonstrate line wrapping and text formatting during export.

One morning, when Gregor Samsa woke from troubled dreams, he found himself transformed in his bed into a horrible vermin. He lay on his armour-like back, and if he lifted his head a little he could see his brown belly, slightly domed and divided by arches into stiff sections. The bedding was hardly able to cover it and seemed ready to slide off any moment. His many legs, pitifully thin compared with the size of the rest of him, waved about helplessly as he looked.

1.2. Lists

Org mode provides several types of lists that can be freely combined to organize information clearly and consistently.

1.2.1 Unordered lists

Bulleted lists are useful for presenting related information without implying a particular order.

They are commonly used to:

- organize information, and
- present content
 - hierarchically, and
 - easy to read.

You can begin list items with either a minus (-) or a plus sign (+).

1.2.2 Ordered lists

Numbered lists express sequences or ordered procedures.

1. First item
 - (a) First sub-item
 - (b) Second sub-item
2. Second item

You can also specify explicit numbering when necessary:

1. First item
2. Second item
5. Continue with item 5

1.2.3 Definition lists

Definition lists associate terms with descriptions.

Definition list A list in which each item consists of a term followed by its definition.

Org mode A powerful plain-text authoring system built into Emacs.

1.2.4 Checklists

Checkboxes make it easy to track progress.

- ☐ Incomplete task
- ☐ Task in progress
 - ☒ First subtask
 - ☐ Second subtask
- ☒ Completed task

1.3. Additional features

1.3.1 Including other Org files

You can include the contents of another Org file during export. For example, the following directive imports the referenced file while skipping its title:

```
#+INCLUDE: chapter1.org :lines "2-"
```

Note

File inclusion using the `#+INCLUDE` keyword is supported **during export only**.

1.3.2 Inline HTML

Raw HTML can be embedded directly in an Org document. When exporting to HTML, the markup is preserved exactly as written.

This text appears in a fixed-width font only in HTML. Embedding raw HTML is particularly useful when you need functionality beyond Org mode's built-in export options, such as custom markup, advanced styling, or interactive content.

For example, you can define custom HTML containers that correspond to CSS classes:

```
#+begin_info
*Info example* \\
Additional details appear here.
#+end_info
```

You may also embed JavaScript, interactive visualizations (such as interactive R plots), or other HTML elements when appropriate.

To include an existing HTML file verbatim during export:

```
#+INCLUDE: file.html html
```

Changes should always be made to the original Org source rather than the exported HTML output!

1.3.3 Inline $\LaTeX$

Raw $\LaTeX$ may also be embedded directly in an Org document.

This text is displayed in a fixed-width font only in $\LaTeX$ { }.

1.3.4 Centered text

Use a `center` block to center its contents.

This text is centered.

1.4. Code blocks

Source code blocks are one of Org mode's most powerful features. They support syntax highlighting, executable code through Org Babel, export to multiple formats, and **reproducible research** workflows.

1.4.1 Diff

Unified diffs are useful for illustrating changes between two versions of a file.

```
diff --git a/config.org b/config.org
index 1a2b3c4..5d6e7f8 100644
--- a/config.org
+++ b/config.org
@@ -1,6 +1,7 @@
  #+TITLE: Example document
  #+AUTHOR: Alice
  +#+LANGUAGE: en

  -* Introduction
  +* Getting started

  -This is the original paragraph.
  +This is the updated paragraph with clearer wording.
```

1.4.2 $\LaTeX$

```
\documentclass[a4paper,11pt]{article}

\usepackage[T1]{fontenc}
\usepackage{amsmath,amssymb}
\usepackage{graphicx}
\usepackage{xcolor}
\usepackage{hyperref}

\title{Syntax Highlighting Demo}
\author{Fabrice Niessen}
\date{\today}
```



```

\begin{document}

\maketitle

\section{Introduction}

This is a simple demonstration of \LaTeX{} syntax.

Inline math:  $(e^{i\pi} + 1 = 0)$ .

Displayed equation:

\[
\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}.
\]

A numbered equation:

\begin{equation}
\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}.
\end{equation}

\subsection{Lists}

\begin{itemize}
\item First item
\item Second item with \textbf{bold} and \emph{emphasis}
\item A colored word: \textcolor{blue}{blue}
\end{itemize}

\subsection{Table}

\begin{tabular}{lcr}
\hline
Name & Value & Unit \\
\hline
Speed & 42 & m/s \\
Mass & 5.3 & kg \\
\hline
\end{tabular}

\subsection{Figure}

\begin{figure}[h]
\centering
\includegraphics[width=0.4\textwidth]{example-image}
\caption{An example image.}
\end{figure}

A hyperlink: \url{https://www.latex-project.org}

\end{document}

```

1.4.3 Line numbers

Both `example` and `src` blocks support optional line numbering. Add the `-n` switch to the `#+begin` line to display line numbers.

```

(defun org-xor (a b)
  "Exclusive or."

```

To continue numbering from a previous numbered block, use the `+n` switch.

```

(if a (not b) b))

```

Literal `example` blocks can also define labels such as `(ref:name)`. These labels become reference targets that may be linked elsewhere in the document, using special hyperlinks like `[[ (name) ]]` (i.e., with the reference name enclosed in single parenthesis).

(In HTML, hovering the mouse over such a link will remote-highlight the corresponding code line, which is kind of cool.)

Use the `-r` switch to suppress labels in the exported source while still allowing references.

With the `-n` switch, links to these references will be labeled by the line numbers from the code listing, otherwise links will use the labels with no parentheses.

```
(save-excursion          ;
  (goto-char (point-min))) ;
```

In line 1, the current cursor position is saved. Line (jump) moves directly to `point-min`.

1.4.4 Program output

One of Org mode's most powerful features is the ability to **execute** code and insert its output directly into the document. This enables *reproducible research*.

Text output

A simple example:

```
date +"%Y-%m-%d"
```

2026-08-12

Longer command output may also be inserted into the document.

```
# Output all styles, but the default one (if any).
ls ../src/ | grep -v "default"
```

```
bigblow
readtheorg
vendor
```

Graphics

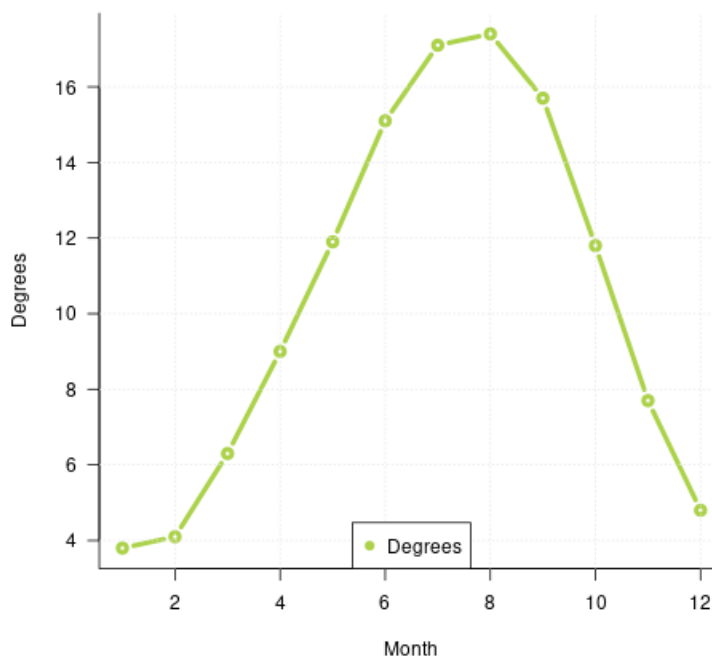
Data to be plotted:

Month	Degrees
1	3.8
2	4.1
3	6.3
4	9.0
5	11.9
6	15.1
7	17.1
8	17.4
9	15.7
10	11.8
11	7.7
12	4.8

The following R code generates a plot from the table above.

```
plot(data, type = "b", bty = "n", col = c("#ABD249"), lwd = 4, las = 1)
grid(nx = NULL, ny = NULL, col = c("#E8E8E8"), lwd = 1)
legend("bottom", legend = c("Degrees"), col = c("#ABD249"), pch = c(19))
```

Generated figure:



R code example

```
library(ggplot2)
summary(cars)
```

Example plot:

```
library(ggplot2)
qplot(speed, dist, data = cars) + geom_smooth()
```

1.5. Inline code

Inline source blocks can be evaluated directly within normal text.

For example, `1 + 1` evaluates to `.`

1.6. Text formatting

1.6.1 Emphasis

Org mode supports several forms of inline emphasis.

Examples include:

- *italic* for emphasis, book titles, or introducing new terms.
- **bold** for strong emphasis or important information.
- ***bold italic*** for very strong emphasis.
- underlined for text that should appear underlined, such as headings or terms that require visual emphasis.
- `inline code` for program code such as `printf("hi")`, commands such as `git status`, variable names, function names such as `strlen()`, or language keywords.
- `verbatim text` for exact strings such as YES, command-line options such as `--force-with-lease`, filenames such as `README.org`, or any text that should be reproduced exactly.
- ~~deleted text~~ for text that has been removed, is obsolete, or should be ignored.
- superscripts such as m/s^2
- subscripts such as H_2O

Markup recognition rules

Formatting can be nested.

For example, *italic text containing underlined text within it*.

Inline markup must be surrounded by appropriate boundaries and cannot be embedded within a word.

For example, `foo*bold*bar` is not recognized because the markup is embedded within a word.

By default, emphasis markup cannot span more than **2 consecutive lines**. For example, the following is not recognized as bold text:

This is not bold

1.6.2 Quotations

Use a `quote` block for quoted prose.

Let us change our traditional attitude to the construction of programs: Instead of imagining that our main task is to instruct a computer what to do, let us concentrate rather on explaining to human beings what we want a computer to do.

The practitioner of literate programming can be regarded as an essayist, whose main concern is with exposition and excellence of style. Such an author, with thesaurus in hand, chooses the names of variables carefully and explains what each variable means. He or she strives for a program that is comprehensible because its concepts have been introduced in an order that is best for human understanding, using a mixture of formal and informal methods that reinforce each other.

— Donald Knuth

Short quotations are equally straightforward.

Everything should be made as simple as possible, but not any simpler. – Albert Einstein

A `verse` block preserves indentation and line breaks exactly as written.

Everything should be made as simple as possible,
but not any simpler. – Albert Einstein

Verse blocks are also useful for preserving the formatting of email conversations.

» This is an email containing nested quotations.
» Each level of indentation is preserved.
>
> Replies remain visually distinct from the original message.

Bullet lists may also appear inside verse blocks:

- First item
- Second item

Numbered lists are equally valid:

1. First item
2. Second item

Maybe an equation here?

See <https://www.google.com/> for additional information...

1.6.3 Spaces

Non-breaking spaces may be inserted using the Unicode character `U+00A0` or an appropriate Org entity, depending on your export requirements.

1.7. Mathematical expressions

Org mode supports $\text{\LaTeX}$ mathematics in both HTML (via MathJax) and $\text{\LaTeX}$ /PDF exports.

- For **inline mathematics**, use `\(...\)`: x^2 or $1 < 2$.
Although `$...$` is commonly used in $\text{\LaTeX}$, `\(...\)` is preferred in Org documents.
- Display equations appear centered on their own line (the *Euler theorem*):

$$\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

Display math using `\[...\]` is useful whenever an equation deserves **visual emphasis**.

For example, the *quadratic formula* may be written as:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Double dollar signs (\$\$) should not be used.

- For **numbered** equations, use the `equation` environment. The *sinus theorem* can then be written as the equation:

$$\frac{\sin \alpha}{a} = \frac{\sin \beta}{b} \quad (1.1)$$

- Equation references are also supported.
See Equation [1.2](#).

$$n_{i+1} = \frac{n_i(d-i)(e-1)}{(i+1)} \quad (1.2)$$

Only captioned equations receive automatic numbering.

- If numbering is not required, use `equation*`, `displaymath`, or the shorter `\[...\]` syntax.

1.8. Special characters

Org mode provides convenient entities for many commonly used symbols.

1.8.1 Accented characters

À Á

1.8.2 Punctuation

Dash: – —

Inverted punctuation: ¡ ¿

Quotation marks: « »

Other symbols: ¶ ^a

1.8.3 Commercial symbols

Copyright: ©

Registered trademark: ®

Currency: ¢ € ¥ £

1.8.4 Greek letters

The Greek letters α , β , and γ are used to denote angles in mathematics.

1.8.5 Mathematical symbols

Arithmetic: ± ÷

Arrows: → ↗ ← ↖ ⇒ ⇐ ⇔

Functions: `arccos` `cos`

Additional symbols: `•` `★`

1.8.6 Miscellaneous symbols

Card suits: `♣` `♠`

1.9. Comments

Comments may be placed anywhere in an Org document.

Lines beginning with `#` are ignored during export.

1.10. Tables

Org mode makes it easy to create and maintain tables.

Table 1.1: Example table

Header 1	Header 2	Header 3
Top left	Top middle	Right
Bottom left	Bottom middle	

Columns are aligned automatically.

- Numeric columns are right-aligned.
- Text columns are left-aligned.

Alignment can also be specified explicitly.

Table 1.2: Table with explicit alignment

1	2	3
right	center	left
XXXXXXXXXXXX	XXXXXXXXXXXX	XXXXXXXXXXXX

Placement may be controlled during $\text{\LaTeX}$ export.

a	b
1	2

This differs from the default centered table.

a	b
1	2

1.10.1 Table alignment on the page

Left-aligned table:

a	b	c
1	2	3
4	5	6

The `noindent` just gets rid of the indentation of the first line of a paragraph which in this case is the table, and the `hfill` adds infinite stretch after the table, so it pushes the table to the left.

Centered table:

a	b	c
1	2	3
4	5	6

Right-aligned table:

a	b	c
1	2	3
4	5	6

Here, the `hfill` adds infinite stretch before the table, so it pushes the table to the right.

1.10.2 Long tables

Long tables may span multiple pages when exporting to PDF. Depending on the $\text{\LaTeX}$ configuration, table headers can automatically be repeated at the top of each page.

1.11. Images, video, and audio

1.11.1 Images

Org mode supports several image formats.

Format	HTML	PDF
GIF	Yes	
JPEG	Yes	
PNG	Yes	
BMP	Browser dependent	

Example image:



Figure 1.1: Org mode logo

Table 1.3: Example of a long table

Row	Project	Owner	Status	Priority	Estimate
1	Alpha	Alice	Done	High	2 d
2	Bravo	Bob	In Progress	Medium	5 d
3	Charlie	Carol	Planned	Low	3 d
4	Delta	David	Done	High	1 d
5	Echo	Emma	Planned	Medium	4 d
6	Foxtrot	Frank	Done	High	2 d
7	Golf	Grace	Planned	Low	8 d
8	Hotel	Henry	Done	Medium	3 d
9	India	Irene	In Progress	High	6 d
10	Juliet	John	Planned	Low	7 d
11	Kilo	Kate	Done	High	2 d
12	Lima	Luke	In Progress	Medium	5 d
13	Mike	Mary	Planned	Low	3 d
14	November	Neil	Done	High	1 d
15	Oscar	Olivia	Planned	Medium	4 d
16	Papa	Peter	Done	High	2 d
17	Quebec	Quinn	Planned	Low	8 d
18	Romeo	Rose	Done	Medium	3 d
19	Sierra	Steve	In Progress	High	6 d
20	Tango	Tina	Planned	Low	7 d
21	Uniform	Uma	Done	High	2 d
22	Victor	Victor	In Progress	Medium	5 d
23	Whiskey	Wendy	Planned	Low	3 d
24	Xray	Xavier	Done	High	1 d
25	Yankee	Yvonne	Planned	Medium	4 d
26	Zulu	Zach	Done	High	2 d
27	Alpha	Alice	Planned	Low	8 d
28	Bravo	Bob	Done	Medium	3 d
29	Charlie	Carol	In Progress	High	6 d
30	Delta	David	Planned	Low	7 d
31	Echo	Emma	Done	High	2 d
32	Foxtrot	Frank	In Progress	Medium	5 d
33	Golf	Grace	Planned	Low	3 d
34	Hotel	Henry	Done	High	1 d
35	India	Irene	Planned	Medium	4 d
36	Juliet	John	Done	High	2 d
37	Kilo	Kate	Planned	Low	8 d
38	Lima	Luke	Done	Medium	3 d
39	Mike	Mary	In Progress	High	6 d
40	November	Neil	Planned	Low	7 d
41	Oscar	Olivia	Done	High	2 d
42	Papa	Peter	In Progress	Medium	5 d
43	Quebec	Quinn	Planned	Low	3 d
44	Romeo	Rose	Done	High	1 d
45	Sierra	Steve	Planned	Medium	4 d
46	Tango	Tina	Done	High	2 d
47	Uniform	Uma	Planned	Low	8 d
48	Victor	Victor	Done	Medium	3 d
49	Whiskey	Wendy	In Progress	High	6 d
50	Xray	Xavier	Planned	Low	7 d
51	Yankee	Yvonne	Done	High	2 d
52	Zulu	Zach	In Progress	Medium	5 d
53	Alpha	Alice	Planned	Low	3 d

You may also link directly to an image file.

[Org mode unicorn image¹](#)

Common HTML image attributes include:

- `align`
- `border`
- `bordercolor`
- `hspace`
- `vspace`
- `width`
- `height`
- `title`
- `alt`

Multiple images can also be arranged side by side using appropriate export options.

1.11.2 Video

Video files are typically referenced through hyperlinks.

A common approach is to use a thumbnail image that links to the video.

`\includegraphics [width=.75\linewidth ]{../images/l-hypnose-c-est-du-fake.png}`²

1.11.3 Audio

Audio resources may be linked using standard Org hyperlinks.

1.12. Special text boxes

Org mode supports several block types for emphasizing content.

1.12.1 Inline tasks

Inline tasks can be used to record reminders or small tasks directly within a document. They may include a TODO keyword, a priority, and tags.

An inline task with a description:

TODO Write the introduction

Draft the introductory section before the next review meeting.

A compact inline task without a description:

¹ [../images/org-mode-unicorn.png](#)

² <https://www.youtube.com/shorts/ev1d8wowPuM>

WAIT `#A` Contact the customer `:phone:`

1.12.2 Example

Example blocks display literal text exactly as written.

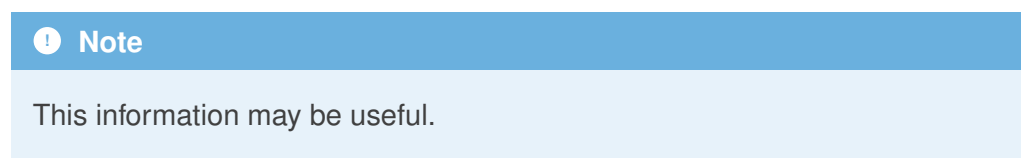
Find entries containing an **exact phrase** by enclosing the phrase in quotation marks:

"hd ready"

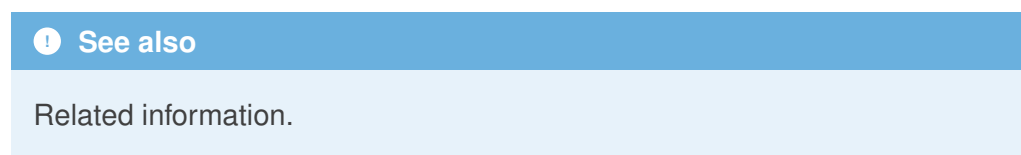
Additional block types include `note`, `seealso`, `tip` and `warning`.

1.12.3 Note and See also

A note box is displayed as follows:

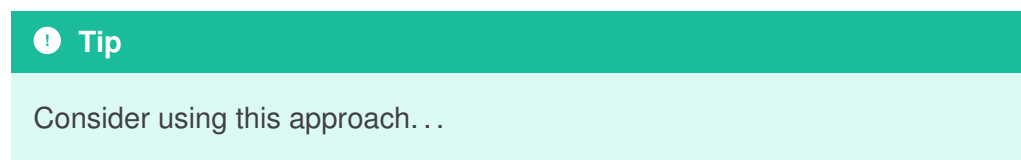


A see-also box is displayed as follows:



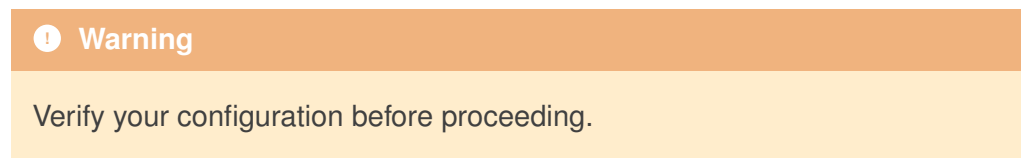
1.12.4 Tip

A tip box is displayed as follows:



1.12.5 Warning

A warning box is displayed as follows:



1.13. Links

1.13.1 Anchors

Org mode supports several types of links.

Most links point to headings, but they may also reference custom anchors within the current document or another Org document.

You can create an anchor using `<<name-of-anchor-here>>` .

1.13.2 Hyperlinks

This document is available in multiple formats:

- [Plain text](#)³
- [HTML](#)⁴
- [PDF](#)⁵

Links are enclosed in square brackets.

Internal links

Examples:

- Chapter [Links](#)
- Section [Anchors](#)
- [Named anchor](#)

External links

Visit the official [Org mode website](#)⁶ .

Email links are also supported.

[Send email](#)⁷

³[example.txt](#)

⁴[example.html](#)

⁵[example.pdf](#)

⁶<https://orgmode.org/>

⁷<mailto:concat.fni.at-sign.pirilampo.org>

2. Org mode features

2.1. Dates

Org mode recognizes both inactive and active timestamps.

Inactive timestamp: *[2014-01-16 Thu]*

Active timestamp: *<2014-01-16 Thu>*

2.2. **DONE** #A Buy GTD book

ONLINE

Completed tasks are typically collapsed by default in exported documents.

2.3. **TODO** #A Read GTD book

SCHEDULED: *<2014-09-11 Thu>*

Scheduled tasks remain expanded unless configured otherwise.

To collapse all headings when exporting HTML with BigBlow, include:

```
#+HTML_HEAD: <script> var HS_STARTUP_FOLDED = true; </script>
```

into your Org document.

2.4. **TODO** #B Apply the GTD methodology

DEADLINE: *<2014-12-01 Mon>*

The `hsCollapsed` class causes this section to appear collapsed when the exported HTML page is first loaded.

Powerful, no?

2.5. Personal note

COMPUTER:WRITE

Tags may be assigned to headings to improve organization and searching.

Interactive HTML exports (such as BigBlow) can also highlight entries that share the same tag.

2.6. Weekly review

COMPUTER

The BigBlow HTML export supports a “review mode”...

Press the `r` key to collapse every active heading except the current one, making it easier to focus on a single task during a review session!

3. Org macros

Macros allow reusable snippets to be defined once and expanded throughout the document.

Example:

This text is displayed in blue.

This other text is displayed in red.

Additional macros are available on [GitHub](https://github.com/fniessen/org-macros)¹.

¹ <https://github.com/fniessen/org-macros>

4. BigBlow extensions

Some HTML themes provide additional features.

For example, the keyword `FIXME` may automatically be replaced with a visual “Fix Me!” indicator:

`FIXME` Delete this. . .